Question 2

1. **Difference-in-Differences function**:

$$\widehat{\delta_{DiD}} = \left[E(Y_i^1 \middle| D_i = 1) - E(Y_i^0 \middle| D_i = 1)\right] - \left[E(Y_i^1 \middle| D_i = 0) - E(Y_i^0 \middle| D_i = 0)\right]$$

2. **Key assumption:**

   In order to make the difference-in-differences estimator an unbiased estimator, the key assumption is the Parallel Trends Assumption, which is the ideal assumption that without the policy intervention, the trend in consumption change should be the same for the intervention and control groups.

   More specifically, it is in the absence of policy intervention, the trend of consumption change should be the same for the intervention group (Gansu Province) and the control group (other provinces).

   It can also be presented in the following equation:

$$E(Y_i^1 - Y_i^0 \mid D_i = 1) = E(Y_i^1 - Y_i^0 \mid D_i = 0)$$

3. **Explanation:**

   The difference-in-differences estimate will Overestimate the treatment effect.

   The DiD method assumes that differences in consumption changes are fully attributable to policy interventions. However, if households in Gansu Province work harder, leading to an increase in income and consumption. Then, this extra increase is unrelated to the policy intervention but will still be accounted for as an effect of the policy by the DiD estimator, leading to an Overestimation.